

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5111

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I Kala Gowri Shankar (192325109) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Kala Gowri Shankar

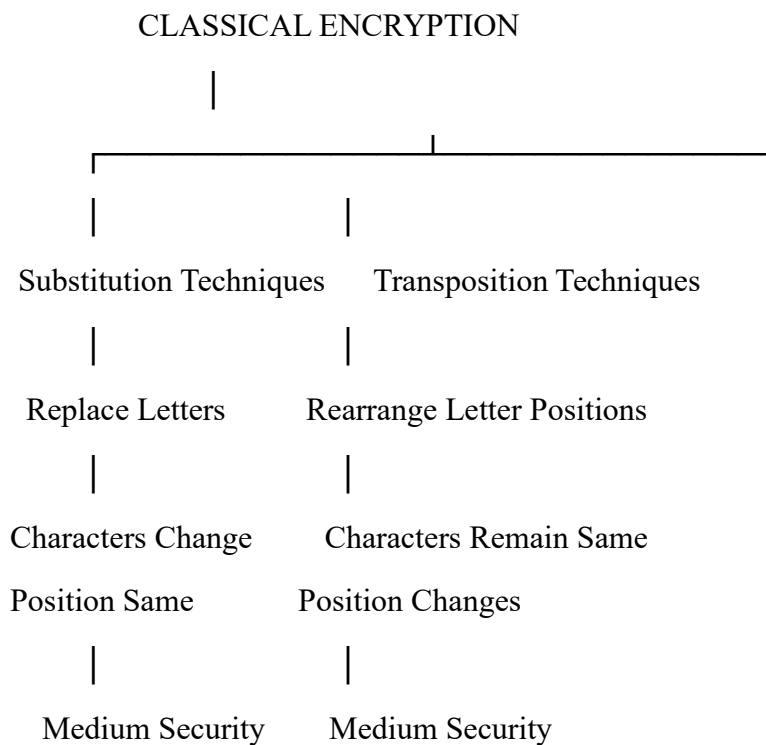
3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:

a. Show differences and similarities

i. Map how each affects the Data structure and Security level

ii. Analyze which is more vulnerable to attacks and why

ANSWER:



(a) Differences and Similarities

Similarities

- Both convert plaintext into ciphertext.
- Both use a secret key.
- Both improve data security.

Differences

Substitution

Replaces letters

Letter values change

Position remains same

Transposition

Rearranges letters

Letter values remain same

Position changes

(i) Data Structure and Security Level

Technique

Data Structure

Security Level

Substitution

Characters are replaced

Medium

Transposition

Character positions are changed

Medium

(ii) Analysis

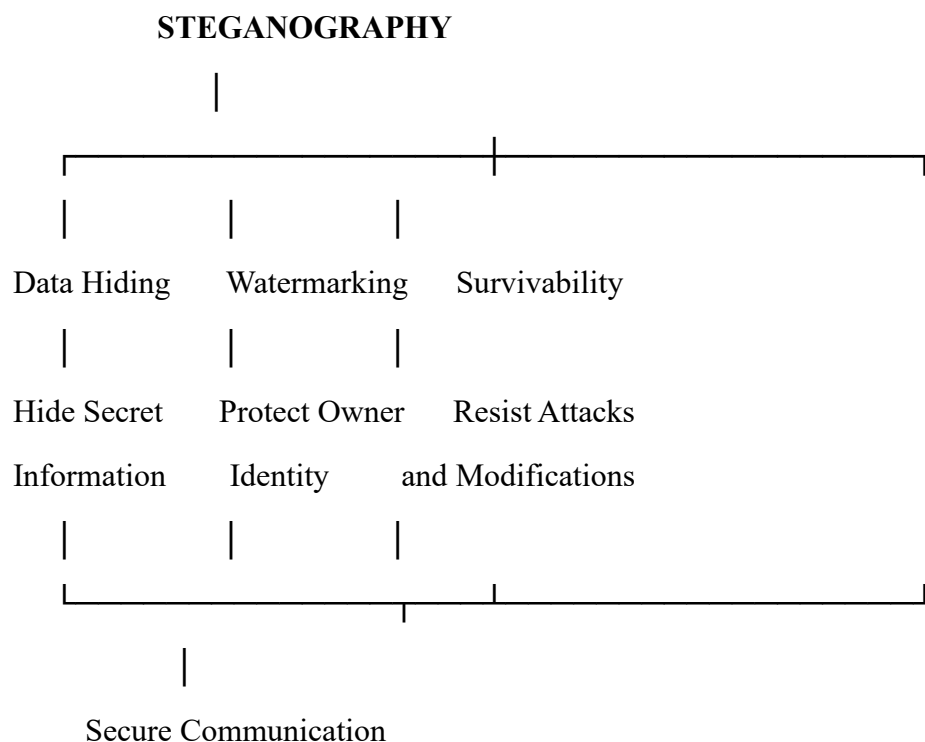
More Vulnerable: Substitution Technique

Reason: It is easier to break using **frequency analysis**, since common letters (E, T, A, etc.) appear frequently in the ciphertext.

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:

- Show relationships between these concepts
- Analyze how each contributes to secure communication
- Include one real-world application for each

ANSWER:



(a) Relationships

- **Data Hiding** hides confidential information inside media.
- **Watermarking** proves ownership of digital content.
- **Survivability** ensures hidden data remains safe even after editing or compression.

(b) Contribution to Secure Communication

Concept	Contribution
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Data Hiding	Keeps messages secret
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Watermarking	Prevents copyright misuse
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Survivability	Protects hidden data from damage or attacks
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(c) Real-World Applications

Concept	Example
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Data Hiding	Secret military communication
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Watermarking	Copyright protection for images and videos
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Survivability	Secure storage of medical images
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Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies.	Good understanding with proper integration of most concepts.	Basic understanding ; limited or partial integration.	Poor understanding; unable to integrate concepts.
Experimental Design	30	Well-planned, systematic implementation with optimal solution.	Correct implementation with minor issues.	Basic implementation with errors or inefficiencies.	Incorrect or incomplete implementation.
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration.	Appropriate use of tools with correct execution.	Limited or inconsistent use of tools.	Incorrect or minimal use of tools.
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results.	Good analysis with reasonable interpretation.	Basic analysis with limited insights.	Poor or incorrect analysis of results.
Documentation	10	Clear, well-structured documentation with diagrams, code, and results.	Organized documentation with necessary details.	Basic documentation with missing details.	Poor or incomplete documentation .